JOVEN PAOLO ANGELES

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EDUCATION

Ph.D in Materials Science and Engineering
University of the Philippines *August 2021 to Present*

- Thesis Title: Bayesian Optimization Framework for Accelerated Nanoparticle Synthesis
- Research adviser: Marienette Vega, Ph.D.
- Focus: Nanoparticle Synthesis, Machine Learning, Computational Material Science, Bayesian Optimization

M.Sc. in Materials Science and Engineering
University of the Philippines *August 2018 to July 2020*

- Thesis Title: Correlation of the Extinction Spectra and SERS Enhancement Factor of Gold Nanostar-Graphene Substrates
- Research adviser: Marienette Vega, Ph.D.
- Focus: Computational Material Science, Finite Element Modelling, Discrete Dipole Approximation, plasmonics, SERS

B.Sc. in Physics
University of the Philippines *June 2011 to June 2015*

- Thesis Title: Fabrication of Triboelectric Nanogenerators
- Research adviser: Ian Jasper Agulo, Ph.D.
- Focus: Material Science, Device Fabrication, Triboelectric nanomaterials

RESEARCH EXPERIENCE

Senior AI Engineer
DOST PROPEL Program
Department of Science and Technology *August 2025 to Present*

- Spearhead the development of Retrieval-Augmented Generation (RAG) pipelines, mainly Agentic RAG and Deep Research Agents, to support commercialization of government-funded R&D outputs.
- Lead multi-agent AI agent architecture, integrating Agno, Ollama, and Postgres for scalable knowledge management across DOST councils.
- Coordinate with cross-functional teams and stakeholders to deliver AI-powered platforms for IP licensing, startup funding, and technology transfer.

Data Scientist
DOST PROPEL Program
Department of Science and Technology *March 2025 to July 2025*

- Built an ELT pipeline for public-sector documents; used LLMs for parsing, deduplication, and data cleaning.
- Modeled and served data in PostgreSQL; enabled hybrid retrieval with `pgvector` for dense embeddings alongside text indexes.
- Exposed a unified databank via WSO2 for downstream analytics and RAG services.
- Implemented data quality checks; automated refresh and backfill jobs for freshness and reproducibility.

University Researcher II
Heterojunction of Graphene-based composites for Membrane Applications in COVID 19 Pandemic Mitigation
Department of Science and Technology *January 2023 to October 2023*

- Develop a graphene-MOF composite
- Handled setup for bulk synthesis of composites through electrospinning

Visiting PhD Student
Kansai-Asia Platform of Advanced Analytical Technologies (KAPLAT) Program
Kyoto University *Oct 2022 to Dec 2022*

- Synthesized gold–iron oxide core–shell nanoparticles for surface-enhanced Raman scattering (SERS) applications.
- Conducted material characterization and calculated SERS enhancement factors.
- Optimized synthesis protocols for stable core–shell nanoparticle structures.
- Applied Raman spectroscopy to investigate magnetoplasmonic properties in exchange-biased ferrofluids.

University Researcher I
NanoSiliCage (NSC): Nanocaged silicate composite for treatment of wastewater from Valenzuela City plastic industries
Department of Science and Technology *January 2022 to December 2022 , November 2023 to December 2023*

- Assess wastewater from recycling companies.

- Modify zeolite/bentonite material for pollutant optimized adsorption.
- Performed efficiency tests on minerals for adsorption of toxins found in industrial wastewater from plastic recyclers.

Science Research Specialist I
Fabrication of Graphene-based Nanostructure Substrates for Applications in Ultrasensitive Detection
Department of Science and Technology *February 2021 to February 2022*

- Fabricated graphene-based hybrid nanostructures for ultrasensitive metal ion detection in water.
- Performed numerical simulations to optimize nanostructures with the highest SERS enhancement factor.

Research Assistant
Anthropological, Mathematical Symmetry and Technical Characterization of Cordillera Textiles - CordiTex Project
Emerging Interdisciplinary Research (EIDR) Program - University of the Philippines *May 2016 to May 2017*

- Characterized mechanical properties of the materials using UTM
- Designed an imaging setup and workflow for faster pattern analysis
- Performed computational analysis on the preserved samples to eliminate the need for destructive characterization methods
- Worked in a highly collaborative project composed of scientist and engineers from different backgrounds
- Prepared administrative works for the research project such as generating monthly progress reports.

Student Researcher
Micro and Nano Innovations Laboratory (MiNILab)
University of the Philippines *April 2014 to July 2014*

- Conducted research on the basic concepts on Graphene, Carbon Nanotubes, and Zinc Oxide nanomaterials.
- Performed quantitative and qualitative analysis of nanomaterials using various characterization methods.
- Performed thin film production methods such as Chemical Vapor Deposition,Electrochemical Deposition, and Spray Pyrolysis Technique.
- Designed and modified the low-cost Spray Pyrolysis setup for more uniform deposition of ZnO nanostructures.

TECHNICAL SKILLS

Computational, AI & Software

- **Programming/Scientific:** Python, MATLAB, R; JupyterLab, Pandas
- **Simulation:** EM modeling (DDA, FEM, FDTD), Density Functional Theory (Quantum ESPRESSO)
- **ML/Optimization:** PyTorch, GPyTorch, BoTorch, Ax, JAX
- **LLMs/RAG & Web/App:** LangChain, OpenAI, Gemini, Ollama, OpenRouter; React, Vite, TypeScript, FastAPI, Flask, Node.js, Streamlit, Plotly, n8n
- **DevOps:** Docker, CI/CD (GitHub Actions, GitLab CI, Jenkins); Observability (Prometheus, Grafana, ELK/EFK, OpenTelemetry); Model serving (vLLM); Experiment tracking (MLflow); Artifacts/Registry (Hugging Face Hub)
- **Blockchain/Web3:** Solidity, Foundry

Experimental Materials Science

- **Microscopy & Imaging:** SEM imaging & analysis, light microscopy
- **Spectroscopy:** Raman spectroscopy, UV–Vis spectroscopy, photoluminescence/fluorescence
- **Crystallography:** XRD characterization
- **Mechanical Testing:** Universal testing machine (UTM) operation
- **Thin Film & Nanomaterial Synthesis:** chemical vapor deposition (CVD), spray pyrolysis
- **Electrochemical Methods:** electrochemical deposition
- **Plasma-Based Processing:** RF plasma synthesis
- **Solution-Based Routes:** sol–gel and related wet-chemical synthesis methods

PROFESSIONAL AFFILIATIONS & ACTIVITIES

NanoSpectroscopy Laboratory **University of the Philippines**
Member *August 2018 to present*
Samahang Pisika ng Pilipinas(Physics Society of the Philippines)
Associate Member *June 2019 to present*
Vacuum Society of the Philippines
Member *February 2020 to present*
Optica(formerly Optical Society of America)
Student Member *October 2019 to June 2025*
Micro and Nano Innovations Laboratory **University of the Philippines**
Member *April 2014 to June 2015*
UP Physics Sphere **University of the Philippines**
Council of Leaders Representative *June 2011 to June 2015*

AWARDS

ASTHRDP-NSC MS Scholar — **Department of Science and Technology**
Full-tuition scholarship with stipend for graduate studies. — *August 2018 to July 2020*

ASTHRDP-NSC PhD Scholar — **Department of Science and Technology**
Full-tuition scholarship with stipend for graduate studies. — *January 2024 - Present*

College Scholar — **University of the Philippines**
For attaining a semester GWA of at least 1.75.

- First Semester AY 2011-2012
- Second Semester AY 2012–2013
- First Semester AY 2014–2015

University Scholar — **University of the Philippines**
For attaining a semester GWA of at least 1.45.

- First Semester AY 2018-2019
- Second Semester AY 2018-2019
- First Semester AY 2019-2020
- Second Semester AY 2021-2022

PUBLICATIONS

Peer-reviewed Journal Publications

1. A. D. S. Montallana, **J. P. D. Angeles**, J. P. Chu, M. P. Sherburne, M. R. Vasquez, and M. Wada, "Fabrication of plasma-reduced silver coupled with titanium dioxide nanoparticles for visible light-driven photocatalysis," *Journal of Vacuum Science Technology B*, vol. 41, no. 4, p. 042204, Jul 2023. https://doi.org/10.1116/6.0002510

Submitted Manuscripts

1. **J. P. D. Angeles** and M. M. Vega, "Correlation of the Extinction Spectra and SERS Enhancement Factor of Gold Nanostar-Graphene Substrates," *Japanese Journal of Applied Physics*, submitted.
2. **J. P. D. Angeles** and M. M. Vega, "Fe-Exchanged Philippine Natural Zeolites for Efficient Nitrate Removal from Aqueous Solutions," *Journal of Environmental Science and Health, Part A*, submitted.
3. G. F. S. Cabrera, **J. P. D. Angeles**, and M. M. Vega, "Synthesis and Ultraviolet-Visible Spectroscopy Studies of Gold@Graphene Oxide Nanocomposites," *Japanese Journal of Applied Physics*, submitted.

In Preparation

1. **J. P. D. Angeles** and M. M. Vega, "Domain-Aware Bayesian Optimization for Tunable Localized Surface Plasmon Resonance in Gold Nanoparticles," *Materials & Design*, in preparation.
2. **J. P. D. Angeles** and M. M. Vega, "One-Pot Synthesis of Gold Nanoparticle-Graphene Sponge Composite," *Journal of Materials Chemistry C*, in preparation.
3. **J. P. D. Angeles** and M. M. Vega, "Bayesian Strategies for Low-Defect Graphene in One-Pot Gold Nanoparticle-Graphene Sponge Synthesis," *Science and Technology of Advanced Materials: Methods*, in preparation.

Book Chapters

1. A. Salvador-Amores, E. Taganap, M. A. de Las Peñas, F. A. Prudente, K. Gutierrez, D. Lucena, J. V. Narvasa, I. J. Agulo, A. T. Garcia, G. M. Malapit, R. Q. Baculi, J. Inovero, R. Kelly, M. Stephens, **J. P. D. Angeles**, M. R. Busacay, J. Pangasinan, N. Garcia, and D. Sabado, "Anthropological Analysis, Mathematical Symmetry and Technical Characterization of Cordillera Textiles," in *CordiTex*, 2019. Available at: `https://corditex.upb.edu.ph/index.php/publications/books/anthropological-analysis/`

CONFERENCE PRESENTATIONS

1. **J. Angeles** and M. Vega, "Leveraging Machine Learning and Plasmonic Graphene Composites for Advanced Detection of Heavy Metals in Water," in *National Research Council of the Philippines (NRCP) Annual Scientific Conference and 91st General Membership Assembly*, Philippine International Convention Center, Pasay City, Metro Manila, Philippines, Mar 2024. **(Best Scientific Poster)**
2. **J. Angeles**, J. Lopez, K. Salazar, and M. Vega, "Exploring the adsorption capacity of acid-modified Philippine Natural Zeolites in wastewater treatment," in *Samahang Pisika ng Pilipinas*, Siargao Island, Surigao del Norte, Philippines, Jul 2023. **(Best Poster Award)**
3. T. Rodriguez, M. Ramos, **J. Angeles**, and M. Vega, "Investigation of water as solvent alternative for methanol-based ZIF-8 synthesis," in *Samahang Pisika ng Pilipinas*, Siargao Island, Surigao del Norte, Philippines, Jul 2023. (Oral Presentation)
4. E. Casalme, R. Andig, **J. Angeles**, and M. Vega, "Synthesis of GO/ and rGO/ZIF-8 heterojunction composites," in *Samahang Pisika ng Pilipinas*, Legazpi City, Albay, Philippines, 2022. [SPP-2022-2E-03] (Poster Presentation)
5. L. Sayson, J. Lopez, **J. Angeles**, M. Tumanguil-Quitoras, and M. Vega, "Removal of methylene blue dye from water using silver-exchanged Philippine natural zeolites," in *Samahang Pisika ng Pilipinas*, Legazpi City, Albay, Philippines, Sep 2022. (Oral Presentation)
6. **J. Angeles** and M. Vega, "SERS Enhancement Factor of Gold Nanoparticle-Graphene Substrates," in *Samahang Pisika ng Pilipinas*, National Institute of Physics (Online), Quezon City, Philippines, Oct 2020. **(Best Scientific Presentation Award)**

7. Y. Rola, **J. Angeles**, G. Cabrera, and M. Vega, "Efficiency Assessment of SERS Hotspots Generated from Gold@Iron Oxide Interface," in *Samahang Pisika ng Pilipinas*, National Institute of Physics, Quezon City, Philippines, Oct 2020. (Poster Presentation)
8. G. Cabrera, **J. Angeles**, Y. Rola, and M. Vega, "Percolated graphene oxide with Au nanosphere spacer for SERS detection of glucose," in *Samahang Pisika ng Pilipinas*, National Institute of Physics, Quezon City, Philippines, Oct 2020. (Poster Presentation)
9. **J. Angeles** and M. Vega, "Surface Plasmon Resonance of $\mathrm{Fe_3O_4}$–Au core-shell nanoparticles for SERS Applications by Finite Element Method," in *National Research Council of the Philippines (NRCP) Annual Scientific Conference and 87th General Membership Assembly*, Philippine International Convention Center, Pasay City, Metro Manila, Philippines, Jun 2020.
10. **J. Angeles** and M. Vega, "Discrete Dipole Approximation for Surface Plasmon Resonance of $\mathrm{Fe_3O_4}$–Au nanoparticles," in *Annual Meeting of the Physical Society of Taiwan*, National Pingtung University, Pingtung, Taiwan, Feb 2020.
11. **J. Angeles**, K. Duque, A. Payot, M. Vasquez, and M. Vega, "Synthesis of Nitrogen-Doped Graphene Oxide using RF Plasma Treatment," in *International Symposium of the Vacuum Society of the Philippines*, Cebu City, Philippines, Jan 2020.
12. **J. Angeles** and M. Vega, "Surface plasmon resonance of $\mathrm{Fe_3O_4}$–Au spherical nanoparticles by discrete dipole approximation," in *ICTP Asian Network School and Workshop on Complex Condensed Matter Systems*, National Institute of Physics, Quezon City, Philippines, Nov 2019.
13. A. Salvador-Amores, et al., "Anthropological, Mathematical Symmetry and Technical Characterization of Cordillera Textiles," in *Office of the Vice President for Academic Affairs Research Symposium*, Quezon City, Philippines, Oct 2016.
14. **J. Angeles** and I. Agulo, "Fabrication of Triboelectric Generators," in *4th Materials of Value and Essence Symposium*, Quezon City, Philippines, Jul 2015. (Poster Presentation)
15. L. Pengson, A. Payot, K. Duque, **J. Angeles**, and I. Agulo, "Voltage–Frequency and Voltage–Surface Area Relationship of Triboelectric Nanogenerator," in *4th Materials of Value and Essence Symposium*, Quezon City, Philippines, Jul 2015. (Poster Presentation)
16. **J. Angeles**, C. Pascua, and I. Agulo, "Surface Modification of Ceramic Through HCl Treatment and its Effect on the Growth of Spray-Pyrolized ZnO," in *Philippine–American Academy of Science Engineering (PAASE) Annual Meeting and Symposium*, De La Salle University, Manila, Philippines, Oct 2014. (Oral Presentation)
17. **J. Angeles** and I. Agulo, "Spray-Pyrolized Zinc Oxide Structures Grown on Surface Modified Ceramic by Acid Treatment," in *3rd Materials of Value and Essence Symposium*, Baguio City, Philippines, Oct 2014. (Poster Presentation)

SKILLS AND CERTIFICATIONS

Civil Service Professional Eligibility — **Philippines**
Certification No. 0029, s. 2016 — *May 2016*

Data Science Specialization (with Distinction) — **John Hopkins University**
May 2016

- The Data Scientist's Toolbox (License no: ZB7NJL28UL73) — *Oct 2017*
- R Programming (License no: 8JNWC386HTF9) — *Oct 2017*
- Getting and Cleaning Data (License no: VT9A8DXVN87F) — *Nov 2017*
- Exploratory Data Analysis (License no: 8CEQ3SGVGASV) — *Dec 2017*

IBM RAG and Agentic AI Specialization (with Distinction) — **IBM**
Certificate no: 96Y3IFM69UFO — *Jul 2025*

- Develop Generative AI Applications (License no: 1UAQXPDHPET7) — *Jul 2025*
- Build RAG Applications (License no: KMBOA6AAH8OT) — *Jul 2025*
- Vector Databases for RAG (License no: C765YI3KHR8N) — *Jul 2025*
- Build Multimodal Generative AI Applications (License no: T4F0NKS9DN9P) — *Jul 2025*
- Fundamentals of Building AI Agents (License no: NBUSB1LP04MB) — *Jul 2025*
- AI Agents and Agentic AI with Python & Generative AI (License no: 6RBJDA2FFDN5) — *Jul 2025*
- Advanced RAG with Vector Databases and Retrievers (License no: 10EXCHKWCNOD) — *Aug 2025*
- Agentic AI with LangChain and LangGraph (License no: PML16ZYC9Z2Z) — *Sept 2025*

Blockchain Specialization (with Distinction) — **University at Buffalo & The State University of New York**
Certificate ID: 0KC8CXJGZBAS — *Sept 2025*

- Blockchain Basics (License no: IOI6RESS8A0A) — *Jul 2025*
- Smart Contracts (Certificate ID: FFZ2EAAIVHKH) — *Sept 2025*
- Decentralized Applications (Dapps) (License no: Z1QXSUG67VRK) — *Sept 2025*
- Blockchain Platforms (License no: 0A92LYAQJ1NS) — *Aug 2025*

OTHER SKILLS & INTERESTS

Software	COMSOL Multiphysics, DDSCAT, LaTeX, ImageJ, Microsoft Office Suite, Adobe Photoshop, Adobe Illustrator.
Languages	English (Professional Proficiency), Filipino (Native).
Entrepreneurship	Founder and Owner of **PRINT3D.MNL**, a 3D printing business specializing in customized prints and design solutions.
Photography	Freelance photographer with multiple international awards; portfolio available at https://jovenpaoloangeles.github.io/.
Interests	Photography, Generative Art, 3D Printing.